Module_3

February 1, 2026

1 Week 3 Practice Worksheet: Python Data Visualization

Objective: This worksheet is designed to help you gain hands-on experience with Python data visualization tools and techniques. By replicating a sample visualization provided in Week 3 materials, you will begin to develop fluency with visual representation of data using libraries like Matplotlib and Seaborn.

Instructions: 1. Choose one example from the Week 3 Python data visualization tutorials (e.g., line plot, bar chart, scatter plot, heatmap). 2. Attempt to replicate the example in your local or cloud-based Python environment (e.g., Jupyter Notebook). 3. Answer the questions below and submit this worksheet along with your code and output (as screenshots or notebook file).

1.1 Part 1: Example Attempted

In this project, I will choose the different visualizations from the **"Big Data and Visualization Examples"** with the purpose to explore practice how to handle larger samples of data, instead of traditional visualizations:

First we will do the example of "small" data

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import seaborn as sns

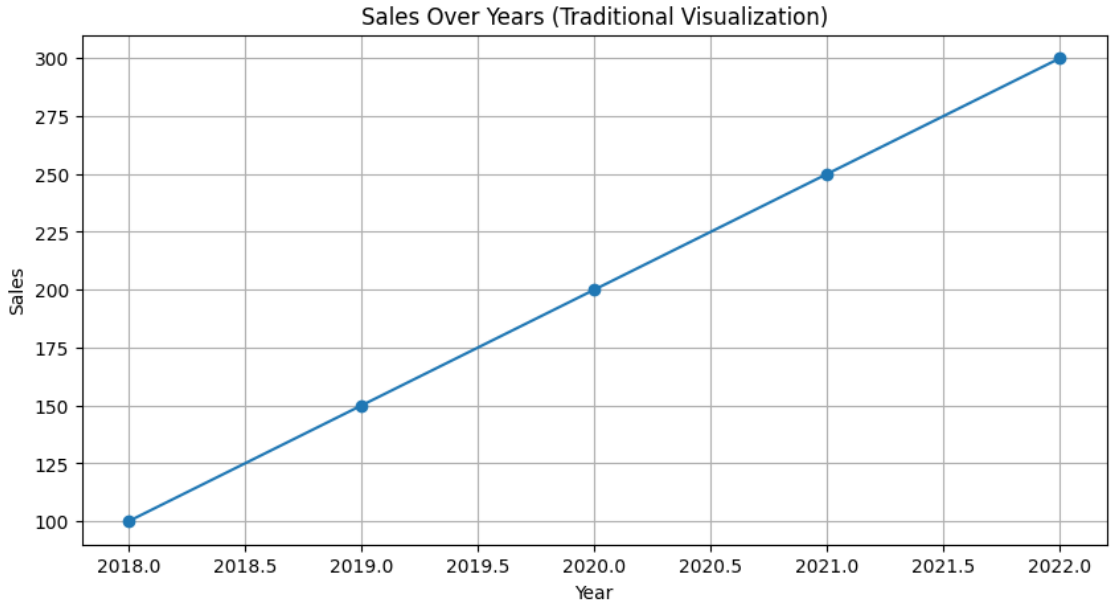

data = {
        "Year":[2018,2019,2020,2021,2022],
        "Sales":[100,150,200,250,300]}

df = pd.DataFrame(data)

#Plotting data

plt.figure(figsize=(10,5))
plt.plot(df['Year'], df['Sales'], marker = "o")
plt.title("Sales Over Years (Traditional Visualization)")
plt.xlabel("Year")
```

```
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



Now I will do the big data example, only that I would make a couple of modifications

[2]:
```
import dask.dataframe as dd

rows = 12**8

larger_data = pd.DataFrame({
        "Year": np.random.randint(2000,2023,size=rows),
        "Sales":np.random.randint(100,1000,size=rows)})

larger_data.to_parquet("data/large_dataset.parquet")
```

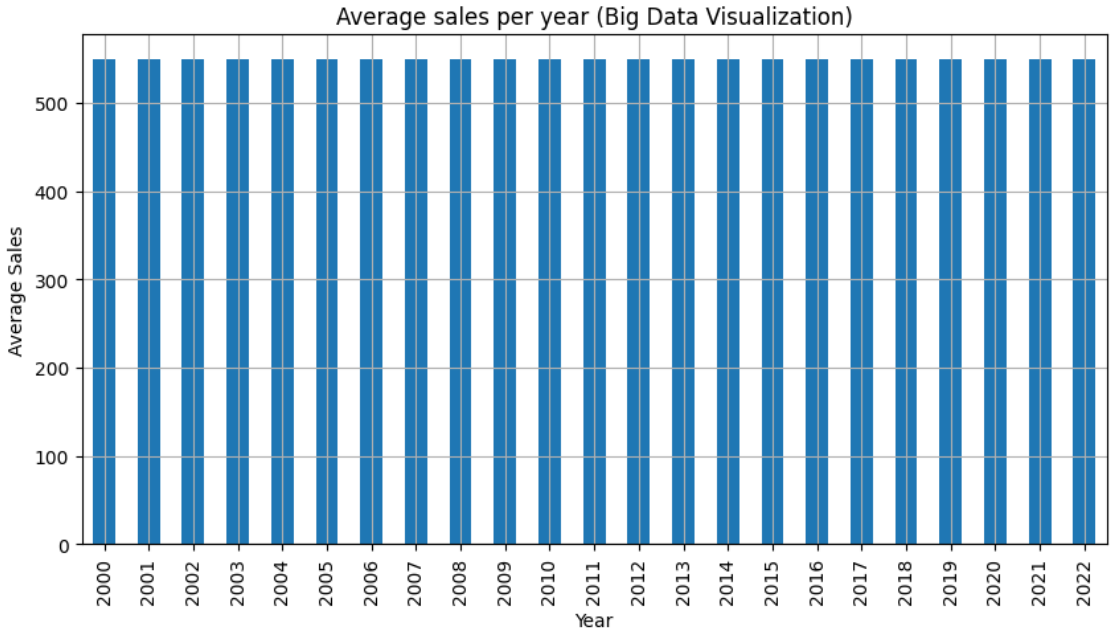
[3]:
```
larger_data = dd.read_parquet("data/large_dataset.parquet")

agg_data = larger_data.groupby("Year")['Sales'].mean().compute()

#Plotting data

plt.figure(figsize=(10,5))
agg_data.plot(kind="bar")
plt.title("Average sales per year (Big Data Visualization)")
plt.xlabel("Year")
plt.ylabel("Average Sales")
```

```
plt.grid(True)
plt.show()
```



1.2 Part 2: Was Your Attempt Successful?

Did your code run without error? Did your output match the example?

As I am familiar with different python libraries, to recreate the example was relatively easy. And also my code generate the exact same result as in the example.

1.3 Part 3: Challenges Encountered

Did you face any issues while attempting the visualization? If so, describe the error(s) or obstacle(s).

To be completly I didn't have any issue while working on this project.

1.4 Part 4: Next Steps

Based on your experience, what would you like to learn or try next? (e.g., customizing plots, trying different chart types, combining plots)

Particularly I would like to explore more customize plots, as in work, we usually do presentations in a certain format, and it has been difficult to adapt matplotlib or seaborn to this format, so at the end I export the aggregate data to Excel. For me that would be useful.

Also to create animation plots might be interesting to improve the story telling of your presentations.